



A Generative AI Powered Approach to Cyberbullying Detection

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CCS CONCEPTS

• **Information Systems** → **Social Networks**; • **Cyber Security** → **Detection**; • **Human-Centered Computing** → **Social Computing**.

KEYWORDS

Cyberbullying, Generative AI, Large Language Model

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1 ABSTRACT

Cyberbullying poses significant risks in online social media environments, leading to severe psychological distress and societal harm. In this paper, we present a systematic approach to cyberbullying detection, aiming to mitigate these risks and protect individuals in digital spaces. Our approach leverages curated datasets from social media platforms and employs preprocessing techniques to extract vital information and calculate confidence scores for indicators of bullying and aggression. We then utilize an open-source Large Language Model (LLM) to generate specialized detection queries based on the extracted data. Responses generated by the LLM are evaluated for signs of cyberbullying using automated classification methods. Specifically, the presence or absence of key indicators within responses determines their classification. Our systematic approach enables the automated identification and categorization of cyberbullying instances, facilitating proactive intervention and prevention strategies. Through this paper, we contribute to the ongoing efforts to combat cyberbullying and promote safer online interactions.

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2 INTRODUCTION

Traditional bullying has been prevalent for a long time in environments like schools [3] [25] and workplaces [16] [24]. The advent of social media has expanded the terrain for bullying, introducing online platforms as new arenas for such behavior [19]. Recent studies indicate that incidents of cyberbullying have escalated over the years, [33] a trend that the COVID-19 pandemic has exacerbated [29]. These developments pose fresh challenges within the technology sector, particularly concerning security [7], as attackers may exploit phishing techniques to hijack victims' accounts and pilfer personal data [18], necessitating robust cybersecurity measures.

Cyberbullying, while occurring in virtual spaces, exacts very real consequences on its victims [8]. It is often linked with significant mental health issues among adolescents, ranging from stress to severe outcomes such as suicide [42] [34]. Efforts to mitigate cyberbullying are thus increasingly focused on enhancing social awareness and bolstering education in cybersecurity [15] [47].

A notable challenge in combating cyberbullying is the scarcity of publicly available datasets [14], which hampers the development and training of effective detection systems. While some studies have attempted to bridge this gap by creating semi-synthetic datasets that encompass various aspects of cyberbullying [14], traditional machine learning methods such as K-Nearest Neighbor (KNN) [31], Support Vector Machine (SVM) [23], Naive Bayes (NB), Decision Trees (DT), and Random Forest (RF) have been the primary tools employed. Although contextual models like BERT have been used, the application of open-source large language models (LLMs) remains underexplored. Our study represents an initial foray into utilizing an open-source LLM, specifically Mistral 7B, to explore its capabilities in detecting cyberbullying. This approach aims to leverage the sophisticated language understanding and processing powers of LLMs to enhance the detection and prevention of cyberbullying in digital communications.

3 RELATED WORKS

Definition of Cyberbullying: The detection of cyberbullying is contingent upon how it is defined, and while there is no universally accepted definition, various academic sources have offered their interpretations. For instance, [20] describes cyberbullying as willful and repeated harm inflicted using computers, cell phones, and other electronic devices. In [36], imbalance of power was also mentioned to be a critical element of cyberbullying. These definitions collectively suggest that for an act to be considered cyberbullying, it must

be intentional, harmful, repetitive, mediated through electronic devices, and involve an imbalance of power between the victim and perpetrators.

Cyberbullying Datasets: Data collection for cyberbullying research has been challenging due to the limited availability of relevant datasets. Twitter, being one of the primary platforms where cyberbullying occurs, has been the focus of several studies. [44] gathered Twitter messages that contained instances of cyberbullying. Instagram API has also been utilized to collect social network data; specifically targeting posts that contained at least one profanity, aiming to filter conversations more likely to involve cyberbullying[21]. Additionally, Ask.fm [39] and Vine [22], [38] social network has also been leveraged to gather data related to cyberbullying.

Cyberbullying Detection Methods: Several methods have been used to detect cyberbullying. Some early studies have used rule based learning to detect cyberbullying [40] [13] [41]. Some studies have used conventional machine learning methods like Support Vector Machines (SVM) [48] [10] [13] [41] [5], Naive Bayes (NB) [13] [5], Logistic Regression (LR) [45] [46] [37], Decision Trees (DT) [9] [12] [13] [41], k-Nearest Neighbours (kNN) [30] [41] and Random Forests (RF) [9] [28]. Lately, deep learning has been used to improve cyberbullying detection. Contextual language models like BERT has also been used to detect cyberbullying [35]. But state-of-the-art large language models are yet to be leveraged for cyberbullying detection.

4 THREAT MODEL

Our threat model as shown in Figure 1, articulates the inherent risks associated with cyberbullying in online social media environments. Users of these platforms engage in various interactions such as sharing images, voicing opinions on contemporary social issues, and responding to feedback on their public posts or private messages. It is within these interactions that the potential for cyberbullying emerges. Cyberbullying is facilitated by the anonymity and lack of immediate consequences typically associated with digital interactions, characterized by persistent personal attacks that cause significant mental distress to the victim [17].

The modus operandi of cyberbullies varies; often, an attacker begins by closely observing a potential victim’s online behavior. Victims perceived as isolated or unsupported are particularly targeted. The harassment may involve attacks based on religion, race, gender, or personal beliefs. In some cases, bullies employ a deceitful approach, initially acting cordially to extract sensitive information such as home addresses, birth dates, and details of other social media accounts. They may also impersonate others, creating fictitious profiles to coerce victims into divulging personal information [6].

Furthermore, cyberbullying is not limited to overtly offensive language or threats; it can also encompass technical attacks [4] such as disabling a victim’s social media presence through mass-reporting, or executing account takeovers via social engineering tactics to publicly disclose personal information [6]. These actions result in severe psychological effects, including stress, isolation, and a withdrawal from social interaction [32], which can suppress free expression and potentially lead to self-harm [27]. The nuanced

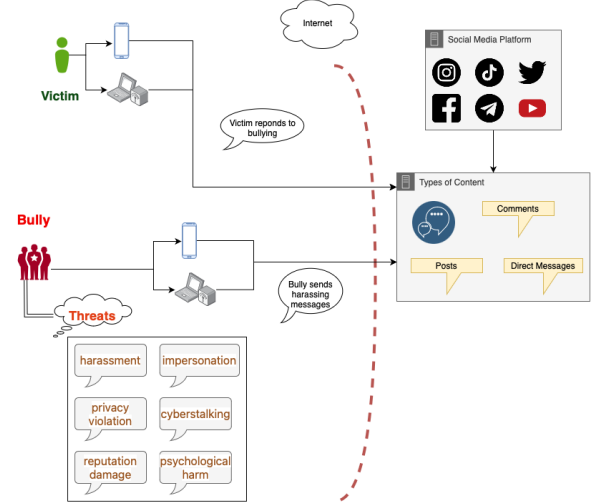


Figure 1: Threat Model

nature of these threats highlights the critical need for robust detection and preventative strategies to protect individuals in digital spaces.

5 SYSTEM MODEL

This section outlines our systematic approach of the design and implementation of our cyberbullying detection system, as depicted in the architecture illustrated in Figure 2 and Algorithm 1. Initially, we curated a dataset by manually gathering instances of cyberbullying from social media platforms such as Instagram dataset [21]. These data files were consolidated into a unified dataset. In the preprocessing phase, we extracted vital information from each conversation within the dataset, including the author, timestamp, and text content. Additionally, we calculated confidence scores for indicators of bullying and aggression, present in the labeled datasets [21]. Each conversation was organized chronologically and subsequently stored in a database.

To facilitate cyberbullying detection, we utilized an open-source Large Language Model (LLM) to create multiple instruction prompts. These prompts were integrated with the data from each conversation to form specialized cyberbullying detection queries. Responses were then generated by the LLM based on these queries and stored in the database. The system evaluates these responses for potential signs of cyberbullying. Specifically, if the word 'yes' is detected within a response, it is classified as indicative of cyberbullying. Conversely, if the word is absent, the response is labeled as not exhibiting cyberbullying. This method allows for a systematic and automated approach to identifying and categorizing instances of cyberbullying within digital communications.

5.1 Data Collection

For our study, we employed an Instagram dataset to evaluate the cyberbullying detection capabilities of an open-source large language model (LLM). The dataset comprises 2,218 Instagram media

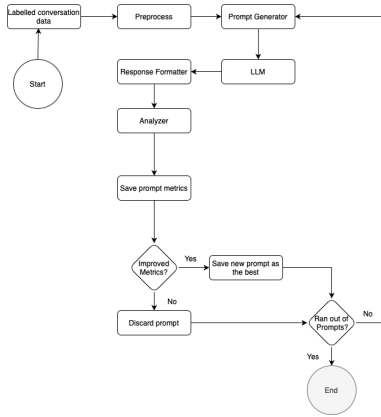


Figure 2: System Model

sessions. It was initially gathered by Hosseinmardi et al. [21] utilizing a snowball sampling method from the Instagram API, which identified 41,000 user IDs. Of these, 25,000 users had public profiles and their media posts and associated comments constituted our sample dataset. Given the high cost of manual labeling, the dataset was refined to include only conversations featuring at least one profanity and a minimum of 15 comments, reducing the size but enhancing the relevance for cyberbullying analysis.

Ultimately, 2,218 media sessions were selected for detailed labeling via CrowdFlower [1], a platform where individuals were trained to identify cyberbullying and cyberaggression. Labelers were provided with specific examples and definitions to differentiate between the two types of content. Quality control was stringent, with only top-performing labelers participating, supplemented by test questions to verify their comprehension of the tasks. Each session was reviewed by five labelers, with safeguards to exclude any participant who did not thoroughly engage with the process [22]. An expert then reviewed the labeled data to confirm accuracy. From the labeled Instagram sessions, 1,954 (88%) exhibited a confidence level greater than 60%. Of these, 29% were classified as ‘bullying’ and 71% as ‘not bullying’. This data will serve as a foundational element for testing the efficacy of our cyberbullying detection system.

5.2 Data Preprocessing

Before employing the datasets for our primary objectives—operating the cyberbullying detection model and evaluating its performance—we undertook a detailed preprocessing phase 2. This involved developing a Python script to transform the datasets, originally in CSV format, into a more model-friendly JSON format, which was then stored in a MongoDB collection for efficient access and manipulation.

Our dataset consisted of CSV files containing Instagram conversations and Vine posts. We initially loaded these files into a pandas DataFrame. The primary focus was on columns such as `_uni_id`, `question1`:confidence, `question2`, and `question2`:confidence. The `_uni_id` served as a unique identifier for each conversation.

Algorithm 1 Cyberbullying Detection Process

```

1: Data collection: Collect labelled Instagram cyberbullying data files  $F$ 
2: Initialize Dataset: dataset  $D \leftarrow$  empty list
3: for each file  $f$  in  $F$  do
4:   Merge in Dataset  $D$ : dataset  $D \leftarrow$  dataset  $D \cup f$ 
5: end for
6: Preprocess:
7: Initialize Database: database  $DB \leftarrow$  empty collection
8: for each conversation  $C$  in  $D$  do
9:   Parse Conversation:
10:  for each comment  $c_i$  in  $C$  do
11:    Remove HTML tags and special characters
12:    Extract author, timestamp, and text
13:  end for
14:  Extract Confidence: Extract bullying confidence, aggression confidence
15:  Sort Comments: Sort comments by timestamp
16:  Save:  $DB \leftarrow DB \cup C$ 
17: end for
18: Generate Prompts:
19: for each  $i$  from 0 to  $n$  do
20:   Generate Cyberbullying Detection Prompt  $P_i$ 
21: end for
22: Response Generation:
23: Initialize LLM
24: for each instruction prompt  $P_i$  do
25:   Retrieve Preprocessed Conversations:
26:   Fetch each conversation from database
27:   for each conversation  $C$  do
28:     Compose Prompt:
29:     Construct prompt with instruction  $P_i$  and conversation  $C$ 
30:     Generate Response:
31:     Utilize LLM to generate response  $r$ 
32:     Save:  $DB \leftarrow DB \cup r$ 
33:   end for
34: end for
35: Detection:
36: Retrieve responses  $R$  from the database  $DB$ 
37: for each response  $r$  do
38:   Convert to Lowercase:  $r \leftarrow \text{lowercase}(r)$ 
39:   if  $r$  contains ‘yes’ then
40:     Save Result:  $DB \leftarrow DB \cup \text{cyberbullying}$ 
41:   else
42:     Save Result:  $DB \leftarrow DB \cup \text{not\_cyberbullying}$ 
43:   end if
44: end for

```

The `question1` column indicated the presence (aggression) or absence (noneAgg) of aggressive content, while `question2` detailed the presence (bullying) or absence (noneBll) of cyberbullying. The confidence columns provided a decimal value between 0 and 1, representing the probability of the conversation containing aggressive or bullying content based on the labelers’ annotations.

Next, we processed the comments data for each conversation. Comments columns were identified by headers beginning with `clmn` or `column`, followed by a numeric suffix that did not necessarily indicate chronological order. We filtered out these columns, discarding any that were empty as part of the data cleaning process. The comments themselves often contained HTML tags, such as those for font colors with CSS, which were parsed and cleaned. We utilized regular expressions to extract the username from the first font color tag within each comment, identifying the author. Additionally, we removed irrelevant HTML tags that could interfere with the detection process. The posting time for each comment, marked by (`created_at`: and concluded with `)`, was also extracted and stored.

After cleaning the text of each comment and removing HTML tags and creation metadata, we sorted the comments within each conversation chronologically. Finally, we constructed a JSON object for each conversation and stored it in our MongoDB collection. This structured approach not only streamlined the dataset but also optimized it for subsequent analysis by our cyberbullying detection model.

Algorithm 2 Data Preprocessing

```

1: Initialize Database: database  $DB \leftarrow$  empty collection
2: for each conversation  $C$  in  $D$  do
3:   Parse Conversation:
4:   for each comment  $c_i$  in  $C$  do
5:     Remove HTML tags and special characters
6:     Extract author, timestamp, and text
7:   end for
8:   Extract Confidence: Extract bullying confidence, aggression confidence
9:   Sort Comments: Sort comments by timestamp
10:  Save:  $DB \leftarrow DB \cup C$ 
11: end for

```

5.3 Language Model

Generative AI are machine learning solutions that are used to generate contents such as texts and images [43]. Language models are probabilistic models of documents.

$$P(w_1, w_2, \dots, w_n) = P(w_1) \times P(w_2 | w_1) \times P(w_3 | w_1, w_2) \times \dots \times P(w_n | w_1, w_2, \dots, w_{n-1}) \quad (1)$$

where $w_1, w_2, \dots, w_n$ are the words in a given sequence and $P(w_i | w_1, w_2, \dots, w_{i-1})$ represents the probability of word w_i given the preceding words in the sequence.

For our cyberbullying detection tasks, we employed the Mistral 7B Instruct v0.1 model, as described in [26]. This model is developed by Mistral AI, a French artificial intelligence company, and contains 7.3 billion parameters, striking a balance between high performance and low computational cost. Mistral 7B employs advanced techniques such as sliding window attention, which enhances its ability to process longer texts efficiently. This feature is crucial for analyzing extensive social media conversations without prohibitive costs. Additionally, the model leverages a rolling

buffer cache to optimize memory usage, and employs a pre-fill and chunking approach for computing attention, both within the cache and across chunks. This methodology allows for more effective sequence generation, significantly outperforming other models with a similar number of parameters. Notably, Mistral 7B surpasses the performance metrics of Llama 2 13B and approaches the coding capabilities of Code-Llama 7B [26].

The Instruct version of Mistral 7B was specifically chosen for our study due to its demonstrated generalization capabilities, having been fine-tuned on publicly available datasets [26]. According to the paper released alongside the model, this version is particularly adept at content moderation tasks. The combination of its smaller size, enhanced efficiency, superior performance, and capability in content moderation renders Mistral 7B Instruct v0.1 an ideal choice for effectively identifying and addressing cyberbullying within digital communications.

5.4 Prompt Generation

As shown in Table 1 specialized prompt generator was implemented to facilitate interaction with the Mistral 7B Instruct model for the purpose of cyberbullying detection. This generator is designed to dynamically create multiple derivative prompts from a single input, adjusting to the context of the conversation at hand. The functionality of the prompt generator is crucial as it enables the model to handle varied inputs and enhances its ability to interpret and respond to nuances within the data. By feeding these contextually adjusted prompts into the Mistral 7B Instruct v0.1 model, we maximize the model's efficacy in detecting instances of cyberbullying.

Input	Output (Mistral 7b instruct v0.1)
I will provide you with an Instagram conversation thread, you will have to detect if it contains cyberbullying or not. Give me an input prompt for a large language model for this task.	Input prompt: "Analyze the following Instagram conversation thread for signs of cyberbullying. If there is any harmful or unethical content, identify and explain it to me."
I will provide you with an Instagram conversation thread, you will have to detect if it contains cyberbullying or not. I need to increase the recall rate of bullying. Give me an input prompt for a large language model for this task.	Input prompt: "Analyze the following Instagram conversation thread for signs of cyberbullying. If there is any harmful or unethical content, identify it in detail and provide a comprehensive explanation of how it can be harmful or damaging to the individuals involved. Additionally, I want you to increase the recall rate of bullying by detecting even subtle examples of bullying that may not explicitly state intent to harm."

Table 1: Sample input and outputs of prompt generation

5.5 Response Generation

Following the creation of dynamic prompts by our prompt generator, we proceeded with the response generation as shown in Table 2 phase using the pre-trained Mistral 7B Instruct model. The initial step involved transforming the input sentences into numeric token IDs, a format the model can interpret. This was accomplished using the AutoTokenizer component of the Hugging Face transformers library. Subsequently, we loaded the pre-trained ‘mistralai/Mistral-7B-Instruct-v0.1’ model from the Hugging Face model hub, utilizing the AutoModelForCausalLM for causal language modeling tasks. To efficiently process the inputs, we employed TextStreamer, which streams text data directly into the model, facilitating smoother and more efficient handling of data. Once the setup was complete, we initialized the model using the Hugging Face pipeline, setting the model’s temperature parameter to 0.001. This low temperature setting was chosen to reduce randomness in the model’s responses, thereby ensuring more deterministic outputs, crucial for maintaining consistency in cyberbullying detection [11].

The prompts were formatted specifically to align with the instructional capabilities of the Mistral 7B model, using the format [INST] prompt [/INST] [2]. In the context of our cyberbullying detection system, each prompt incorporated the user’s comment in a structured template username: comment, which helped in maintaining clarity and relevance in the model’s responses. This structured approach allowed the model to generate targeted responses for each conversation, effectively assessing the presence of cyberbullying within the dialogue.

[INST]prompt[/INST]

We concatenated the comments with the username in a template like this,

{username: comment}

and generated responses for each conversation.

[INST] Analyze the following Instagram conversation thread for signs of cyberbullying. If there is any harmful or unethical content, identify and explain it to me.
{username: comment} [/INST]

Table 2: Sample Prompt for Analyzing Instagram Conversation

5.6 Performance Metrics

To rigorously assess the effectiveness of our cyberbullying detection system, we employed a comprehensive set of metrics: accuracy, bullying precision, non-bullying precision, bullying recall, non-bullying recall, bullying F1-score, and non-bullying F1-score. As shown in Figures 3 and 4, each of these metrics was calculated for individual prompts to ensure nuanced performance evaluation. Our primary focus was on bullying recall, a crucial metric for ensuring that instances of cyberbullying are not overlooked by the system. This metric measures the model’s ability to correctly identify actual instances of bullying among all relevant instances in the dataset.

During our testing process, we iteratively evaluated each prompt for its bullying recall score. If a prompt’s bullying recall exceeded

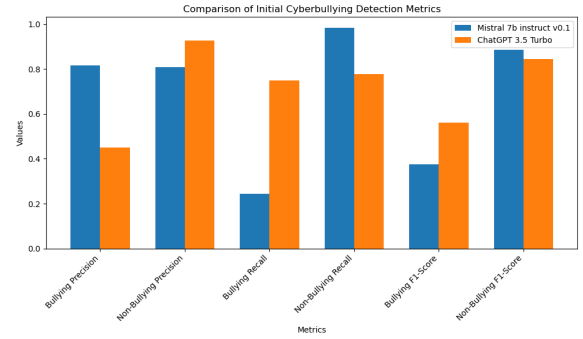


Figure 3: Comparison of Initial Cyberbullying Detection Metrics between Mistral 7b instruct v0.1 and ChatGPT 3.5 Turbo

the highest score recorded up to that point, we identified that prompt as the most effective. This “best prompt” was then saved along with its associated performance metrics in the database for further analysis and future reference. Conversely, prompts that yielded a bullying recall lower than the highest recorded were discarded. This selective approach allowed us to optimize the detection system towards the highest possible sensitivity in identifying bullying instances.

6 RESULTS AND ANALYSIS

In this section, we present the results of our cyberbullying detection experiment. Our initial detection metrics are given in Table 3.

Metric	Mistral 7b instruct v0.1 (Prompt1)	ChatGPT 3.5 Turbo (Prompt1)
Bullying Precision	0.8169	0.4491
Non-Bullying Precision	0.8079	0.9276
Bullying Recall	0.2447	0.75
Non-Bullying Recall	0.9830	0.7767
Bullying F1-Score	0.3766	0.5618
Non-Bullying F1-Score	0.8869	0.8455

Table 3: Comparison of Initial Cyberbullying Detection Metrics between Mistral 7b instruct v0.1 and ChatGPT 3.5 Turbo

As shown in Table 3, the initial evaluation of our cyberbullying detection system using the Mistral 7B Instruct model yielded mixed results. The bullying precision was notably high at 0.8169, indicating that when the model identified conversations as cyberbullying, it did so with a high degree of accuracy. However, the initial bullying recall was considerably lower at 0.2447, signifying that a significant number of bullying instances were missed by the model. Consequently, the bullying F1-score, which balances precision and recall, was also low at 0.3766, underscoring the need for improvement in our model’s sensitivity.

To address this, we embarked on a series of optimizations focused primarily on enhancing the bullying recall. This involved refining our prompt generation process to better capture nuances

Metric	Mistral 7b instruct v0.1 (Prompt1)	Mistral 7b instruct v0.1 (Prompt2)
Bullying Precision	0.8169	0.3871
Non-Bullying Precision	0.8079	0.8244
Bullying Recall	0.2447	0.7788
Non-Bullying Recall	0.9830	0.4571
Bullying F1-Score	0.3766	0.5171
Non-Bullying F1-Score	0.8869	0.5881

Table 4: Comparison of Final Cyberbullying Detection Metrics with Initial Results

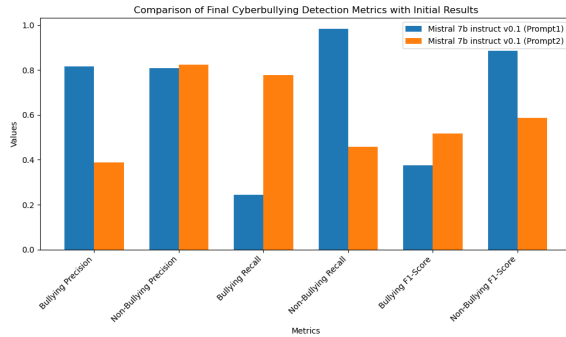


Figure 4: Comparison of Final Cyberbullying Detection Metrics with Initial Results

that indicate bullying behavior. Through iterative testing and adjustments, we managed to significantly improve the bullying recall to 0.7788 which can be seen from Table 4. This improvement in recall was also reflected in the bullying F1-score, which increased to 0.5151 from the initial 0.3766.

While our efforts to boost bullying recall were successful, they did come at the cost of reduced non-bullying recall and F1-score. This trade-off was a calculated decision driven by our prioritization of detecting as many genuine instances of bullying as possible, even if it meant accepting a lower precision. The final iteration of our system demonstrated a refined capability in detecting cyberbullying with significantly improved recall, despite a decrease in precision. This highlights the effectiveness of our targeted approach in prompt engineering, where we chose to prioritize the detection of bullying incidents to better protect individuals from the harms of cyberbullying. Our best performing prompt and its results are given in Table 5.

7 CONCLUSION

In this work, we focused on using Mistral 7b instruct v0.1, an open-source Large Language Model (LLM) to detect cyberbullying. We developed multiple python programs to automate our tasks such as preprocessing Instagram cyberbullying data, initializing model, generating output from the model and then formatting the response for analysis. Our study has been limited to zero shots prompts only.

Best Prompt	Performing Metric	Result
Do the following comments of a conversation contain a repetition of bullying? Here are the comments:	Bullying Precision	0.3871
	Non-Bullying Precision	0.8244
	Bullying Recall	0.7788
	Non-Bullying Recall	0.4571
	Bullying F1-Score	0.5171
	Non-Bullying F1-Score	0.5881

Table 5: Best Performing Prompt and Its Results for Cyberbullying Detection

Our future scopes include few-shots prompting or even fine-tuning LLMs to increase the F1-score of our detection system.

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